

Yasteel Ramphal

Embedded Systems Engineer, Sedna Inc

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📜 Code B Licence

Professional Summary

I am an Embedded Systems Engineer with a BSc Honours (Eng) in Electrical and Information Engineering from the University of the Witwatersrand. I am currently developing custom embedded systems, device drivers, and Linux platforms at Sedna Inc for industrial and consumer product applications.

I bring industrial automation expertise from SAB (AB InBev) including SCADA systems, PLC programming, and process control engineering. I have strong technical depth across embedded hardware, control systems, DSP, machine learning, and telecommunications.

I have proven ability to bridge hardware and software across the full product development lifecycle. I thrive in multidisciplinary R&D environments with a focus on innovation-driven product development.

Skills

Electronic & Embedded Systems

- System architecting, vulnerability detection
- Product lifecycle support, field deployment and maintenance design
- YOCTO, Device drivers (C/Rust), Custom Linux (kernel, device trees, rootfs)
- STM32H7, iMX8, PIC MCU, Raspberry Pi; Board bring-up & UART/SPI/I²C protocols
- Bare-metal MCU programming, FreeRTOS (personal projects & evaluation)
- Telemetry systems, Real-time database monitoring, Grafana
- PCB design, Analogue circuit design & DSP
- Fault tracing, soldering & rework
- Embedded security & crypto accelerators
- LTSpice, KiCAD, Autocad

Industrial Automation

- SCADA (Schneider, Krones, Rockwell)
- PLC programming & logic
- Instrumentation (valves, pumps, sensors)
- Process control (PIC, MPC & FCL)
- SCADA protocols & network monitoring
- LOTO safety procedures, GCC requirements

Education

UNIVERSITY OF THE WITWATERSRAND

BSc Hons (Eng) Electrical and Information Engineering

Courses in control systems, embedded systems, DSP, telecommunications, power systems, machine learning, network and software engineering.

SHANGRI-LA ACADEMY INDEPENDENT SCHOOL

IEB National Senior Certificate

CERTIFICATES & COURSES

AWS Machine Learning Cloud Camp: Rekognition & Textract API, image and text recognition.

HarvardX Cloud Computing — edX: Service models, architecture, security.

Data Science Program Online: R programming, data wrangling, visualisation.

Experience

SEDNA INC

Embedded Systems Engineer — Greenstone, Gauteng **Sept 2025 – Present**

- Architected, developed and deployed a custom embedded telemetry system for real-time data acquisition, processing, and transmission across distributed industrial product platforms
- Writing and maintaining device drivers for custom hardware, interfacing sensors, comms modules, and peripheral ICs on embedded Linux targets
- Building and customising Linux operating systems (kernel configuration, device trees, rootfs) tailored for specific embedded hardware deployments
- Conducting full-lifecycle electronics testing, schematic review, board bring-up, signal integrity analysis, and fault-finding
- Designing system architecture, component selection and integration
- Developed a Linux kernel driver to interface an ignition control peripheral, managing board power sequencing and ensuring safe system shutdown on STM32H7 and iMX8 targets
- Implemented WiFi support on a custom embedded board, integrating the kernel module, driver configuration, and network stack bring-up on an embedded Linux target
- Managed firmware and OS image codebases on GitLab using a branching workflow; CI/CD pipelines automate OS image builds and remote site deployments

SOUTH AFRICAN BREWERIES (SAB – AB INBEV)

Process & Control Engineering Intern — Chamdor Brewery **Dec 2024 – Mar 2025**

- Developed and maintained SCADA systems (Schneider, Krones) for brewery operations, improving real-time production monitoring and process efficiency
- Applied control concepts to optimise manufacturing processes; troubleshoot Schneider and Krones systems for efficiency improvements
- Troubleshoot PLC faults, motor driver issues, and network failures, created PLC system logic and assisted SCADA specialists with virtual client setup
- Familiarised with industrial instrumentation including valves, heat exchangers, pumps, O₂ meters, flow sensors, and pressure transducers
- Designed and commissioned an electrical clean room to ISO standards
- Investigated high power consumption via load analysis and delivered efficiency recommendations
- Contributed to fire safety improvements, elevator lift government compliance, and CIP system fault resolution. Collaborated with cross-functional teams

Software Engineering

- C, C++, C#, Python, SQL, JavaScript, Rust
- Git/GitLab, CMake, BitBake, Cargo, Linux scripting, Docker
- RESTful API design, OAuth 2.0, JWT auth
- Multi-threaded & async programming
- Flask, FastAPI, Node.js backend development
- CI/CD (GitLab pipelines), OS image build & site deployment automation
- PostgreSQL, SQLite, database schema design
- PyTorch, Scikit-learn, Pandas, NumPy
- ETL/ELT pipelines, data warehousing, Power BI
- AWS (Rekognition, EC2, S3, Lambda)
- MATLAB/Simulink,
- Wireshark, GNU Radio, Stable Baselines3

Selected Achievements

- **Van Der Linder Trophy**
Top achiever in Physical Science
- **Math Olympiad**
Qualified SA participant
- **Red Belt in Kung Fu**
Discipline, focus, and perseverance
- **Kaggle Competition**
Titanic ML competition (score: 0.79186)

Community Service

- SPCA Volunteering
- Volunteered at old age homes
- Volunteer at the Food for Life Programme

References

Tafadzwa Raiva
Technical Services Manager, SAB
TRaiva7@gmail.com
074 532 5400

Ashlen Ramsaroop
Process & Control Engineer, SAB
079 693 0441

Sabelo Mazibuko
Manufacturing Systems Manager, SAB
076 986 9104

Joe Neeran
Owner, UVI Tech
084 272 6436

UVI TECH

- Lab Technician** — Pomona Jan – Oct 2024
- Managed manufacturing and assisted with sales by managing client requests; conducted QC testing and spectral analysis of pigments and inks
 - Reported results on manufactured goods, managed logistics

WITS BUSINESS INTELLIGENCE SERVICES

- Data Science Intern** — Johannesburg Jun – Aug 2023
- Created and trained ML models for predictive modelling to support BI
 - Performed EDA, preprocessing, cleaning data and feature extraction
 - Implemented NLP models and REST API integration
 - Managed and coordinated a cross-functional technical team

WITS BUSINESS INTELLIGENCE SERVICES

- Data Engineering Intern** — Johannesburg Sept – Dec 2023
- Leveraged database schemas to design data warehouses and data lakes.
 - Created ETL and ELT processes, feature engineering and data cleaning
 - Integrating cloud services; querying and representing data for business insights through AWS services
 - Team management responsibilities across technical deliverables

MASTER MATHS

- Maths & Physical Science Tutor** — Edenvale 2020
- Monitored and assessed student performance; created productive tutoring environment, communicated lesson topics.

INDEPENDENT

- Assistant Mechanic** 2019
- Diagnosed and troubleshoot mechanical issues; performed routine maintenance and servicing. Ensured compliance with safety procedures

Leadership & Project Management

Technical Team Leadership

- Managed and coordinated cross-functional technical teams during Data Science and Engineering internships, overseeing task allocation, deliverables, and quality assurance
- Lead a team of process engineers, electricians, and maintenance teams at SAB to diagnose faults, implement solutions, and improve plant uptime

Project Delivery & R&D

- Contributing to end-to-end product development at Sedna Inc — from mechanical/electronic design and embedded firmware through system testing and production readiness

Engineering Projects

For detailed reports and source code, visit: <https://yasteelr.github.io/YasteelR/>

UAV-IRS Energy Optimisation via Deep RL

RL framework for joint UAV trajectory and IRS phase shift optimisation in multi-user wireless networks. Continuous RL control integrated with physical-layer channel modelling.

Smart Waste Sorting System

End-to-end waste classification using sensor fusion and on-device ML inference. Hardware actuation with trained models deployed on resource-constrained MCU.

EMG Measurement System for Bionic Hand

Multi-channel signal acquisition for prosthetic hand muscle activation control. Custom analogue hardware design with instrumentation amps, active filtering, and real-time DSP.

Custom Network Synchronisation Protocol

Custom application-layer protocol for time-critical multiplayer synchronisation. Three-phase latency compensation with cryptographic auth over multi-threaded sockets

Secure Computing: LLM Security Evaluation

Benchmarked open-source LLMs on OWASP/CWE vulnerability detection. Prompt engineering with chain-of-thought reasoning for improved security analysis for integration with CI/CD pipelines